

MOSFET – P-Channel, 2.5 V Specified, POWERTRENCH® FDN302P

General Description

This P-Channel 2.5 V specified MOSFET uses a rugged gate version of onsemi's advanced POWERTRENCH process. It has been optimized for power management applications with a wide range of gate drive voltage (2.5 V – 12 V).

Features

- 20 V, –2.4 A $R_{DS(ON)} = 0.055 \Omega @ V_{GS} = -4.5 V$
 $R_{DS(ON)} = 0.080 \Omega @ V_{GS} = -2.5 V$
- Fast Switching Speed
- High Performance Trench Technology for Extremely Low $R_{DS(ON)}$
- SUPERSOT™ –3 Provides Low $R_{DS(ON)}$ and 30% Higher Power Handling Capability than SOT–23 in the Same Footprint
- This is a Pb-Free and Halide Free Device

Applications

- Power Management
- Load Switch
- Battery Protection

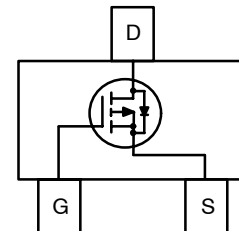
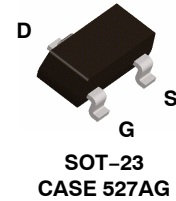
ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ C$ unless otherwise noted.)

Symbol	Parameter	Value	Unit
V_{DSS}	Drain–Source Voltage	–20	V
V_{GSS}	Gate–Source Voltage	± 12	V
I_D	Drain Current – Continuous (Note 1a) – Pulsed	–2.4 –10	A
P_D	Maximum Power Dissipation (Note 1a) (Note 1b)	0.5 0.46	W
T_J, T_{STG}	Operating and Storage Junction Temperature Range	–55 to +150	$^\circ C$

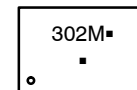
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS ($T_A = 25^\circ C$ unless otherwise noted.)

Symbol	Parameter	Value	Unit
$R_{\theta JA}$	Thermal Resistance, Junction–to–Ambient (Note 1a)	250	$^\circ C/W$
$R_{\theta JC}$	Thermal Resistance, Junction–to–Case (Note 1)	75	$^\circ C/W$



MARKING DIAGRAM



302 = Specific Device Code
M = Month Code
▪ = Pb-Free Package

(Note: Microdot may be in either location)

ORDERING INFORMATION

Device	Package	Shipping†
FDN302P	SOT-23 (Pb-Free/ Halide Free)	3000 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

FDN302P

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
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OFF CHARACTERISTICS

BV_{DSS}	Drain–Source Breakdown Voltage	$V_{GS} = 0\text{ V}, I_D = -250\text{ }\mu\text{A}$	-20	–	–	V
$\frac{\Delta BV_{DSS}}{\Delta T_J}$	Breakdown Voltage Temperature Coefficient	$I_D = -250\text{ }\mu\text{A}$, Referenced to 25°C	–	-12	–	mV/ $^\circ\text{C}$
I_{DSS}	Zero Gate Voltage Drain Current	$V_{DS} = -16\text{ V}, V_{GS} = 0\text{ V}$	–	–	-1	μA
I_{GSSF}	Gate–Body Leakage, Forward	$V_{GS} = 12\text{ V}, V_{DS} = 0\text{ V}$	–	–	100	nA
I_{GSSR}	Gate–Body Leakage, Reverse	$V_{GS} = -12\text{ V}, V_{DS} = 0\text{ V}$	–	–	-100	nA

ON CHARACTERISTICS (Note 2)

$V_{GS(th)}$	Gate Threshold Voltage	$V_{DS} = V_{GS}, I_D = -250\text{ }\mu\text{A}$	-0.6	-1.0	-1.5	V
$\frac{\Delta V_{GS(th)}}{\Delta T_J}$	Gate Threshold Voltage Temperature Coefficient	$I_D = -250\text{ }\mu\text{A}$, Referenced to 25°C	–	3	–	mV/ $^\circ\text{C}$
$R_{DS(on)}$	Static Drain–Source On–Resistance	$V_{GS} = -4.5\text{ V}, I_D = -2.4\text{ A}$ $V_{GS} = -2.5\text{ V}, I_D = -2\text{ A}$ $V_{GS} = -4.5\text{ V}, I_D = -2.4\text{ A}, T_J = 125^\circ\text{C}$	– – –	44 64 58	55 80 84	m Ω
$I_{D(on)}$	On–State Drain Current	$V_{GS} = -4.5\text{ V}, V_{DS} = -5\text{ V}$	-10	–	–	A
g_{FS}	Forward Transconductance	$V_{DS} = -5\text{ V}, I_D = -2.4\text{ A}$	–	10	–	S

DYNAMIC CHARACTERISTICS

C_{iss}	Input Capacitance	$V_{DS} = -10\text{ V}, V_{GS} = 0\text{ V},$ $f = 1.0\text{ MHz}$	–	882	–	pF
C_{oss}	Output Capacitance		–	211	–	pF
C_{rss}	Reverse Transfer Capacitance		–	112	–	pF

SWITCHING CHARACTERISTICS (Note 2)

$t_{d(on)}$	Turn–On Delay Time	$V_{DD} = -10\text{ V}, I_D = -1\text{ A},$ $V_{GS} = -4.5\text{ V}, R_{GEN} = 6\text{ }\Omega$	–	13	23	ns
t_r	Turn–On Rise Time		–	11	20	ns
$t_{d(off)}$	Turn–Off Delay Time		–	25	40	ns
t_f	Turn–Off Fall Time		–	15	27	ns
Q_g	Total Gate Charge	$V_{DS} = -10\text{ V}, I_D = -2.4\text{ A},$ $V_{GS} = -4.5\text{ V}$	–	9	14	nC
Q_{gs}	Gate–Source Charge		–	2	–	nC
Q_{gd}	Gate–Drain Charge		–	3	–	nC

DRAIN–SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS

I _S	Maximum Continuous Drain–Source Diode Forward Current		–	–	–0.42	A
V _{SD}	Drain–Source Diode Forward Voltage	V _{GS} = 0 V, I _S = –0.42 (Note 2)	–	–0.7	–1.2	V

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

NOTES:

- $R_{\theta JA}$ is the sum of the junction–to–case and case–to–ambient thermal resistance where the case thermal reference is defined as the solder mounting surface of the drain pins. $R_{\theta JC}$ is guaranteed by design while $R_{\theta CA}$ is determined by the user's board design.



a) 250°C/W when mounted on a 0.02 in^2 pad of 2 oz. copper.



b) 270°C/W when mounted on a minimum pad.

Scale 1:1 on letter size paper.

- Pulse Test: Pulse Width $\leq 300\text{ }\mu\text{s}$, Duty Cycle $\leq 2.0\%$.

TYPICAL CHARACTERISTICS

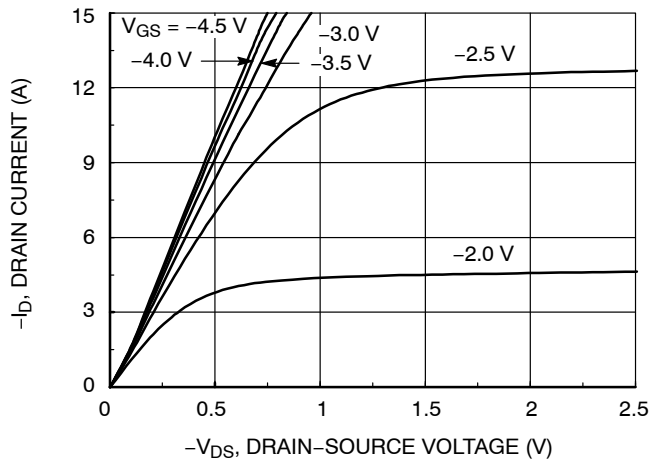


Figure 1. On-Region Characteristics

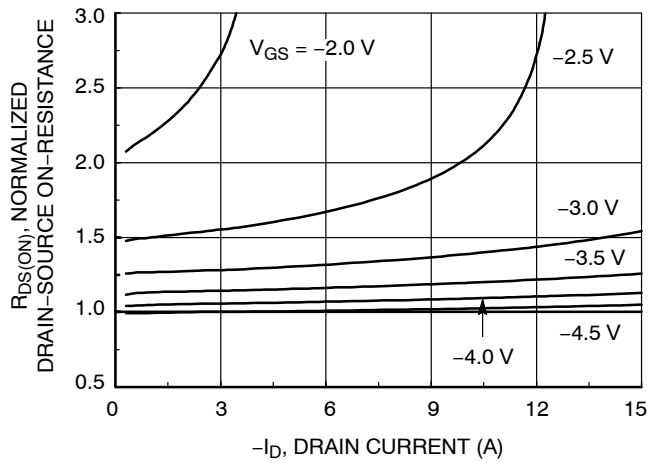


Figure 2. On-Resistance Variation with Drain Current and Gate Voltage

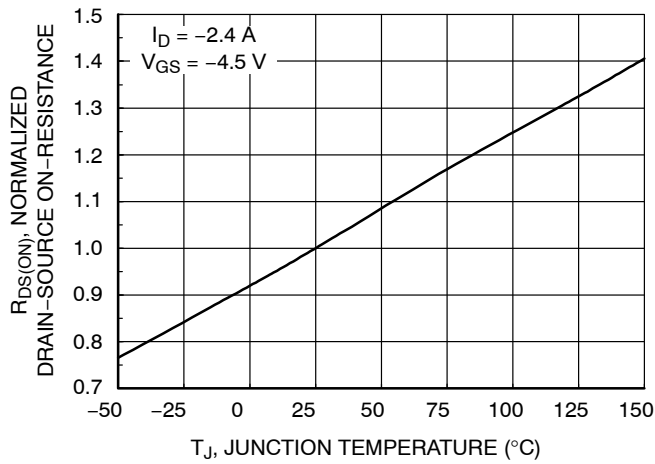


Figure 3. On-Resistance Variation with Temperature

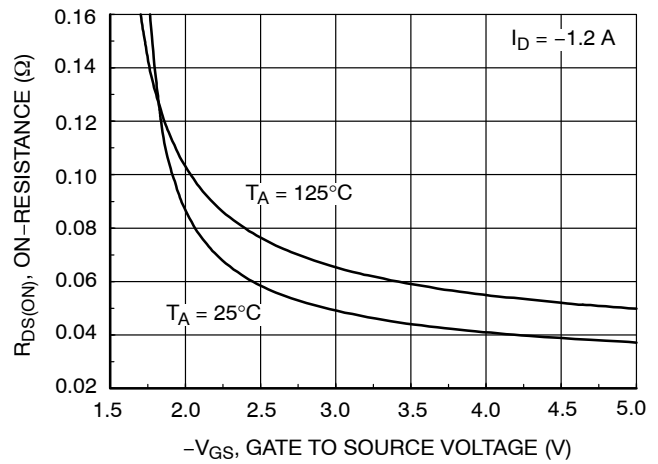


Figure 4. On-Resistance Variation with Gate-to-Source Voltage

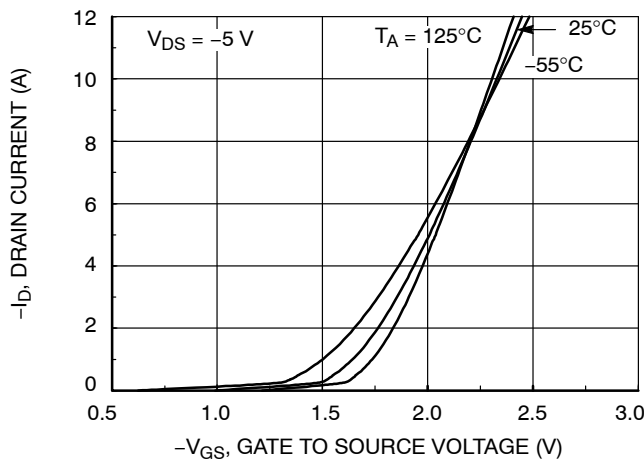


Figure 5. Transfer Characteristics

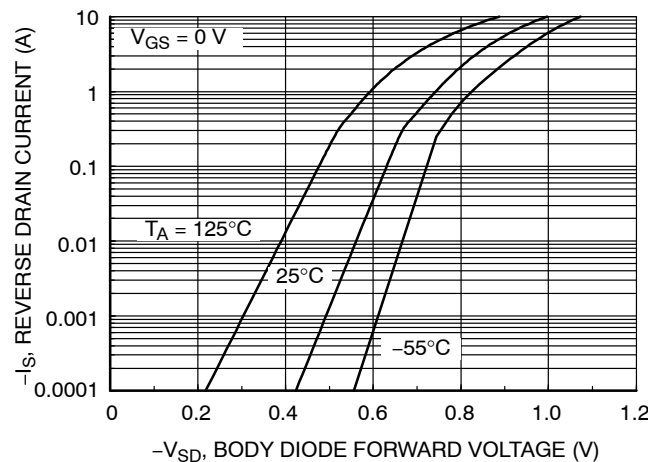


Figure 6. Body Diode Forward Voltage Variation with Source Current and Temperature

TYPICAL CHARACTERISTICS (Continued)

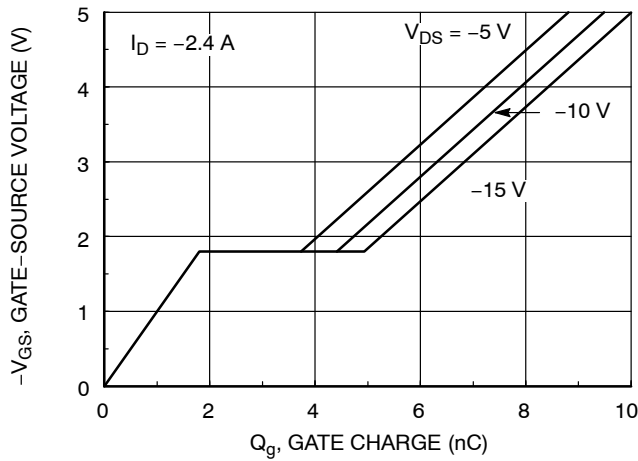


Figure 7. Gate Charge Characteristics

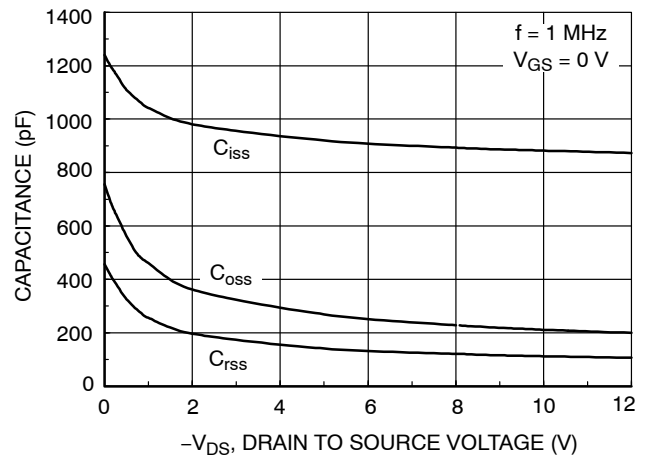


Figure 8. Capacitance Characteristics

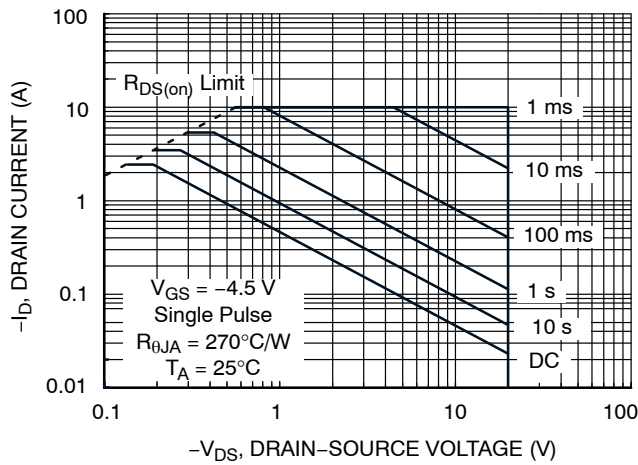


Figure 9. Maximum Safe Operating Area

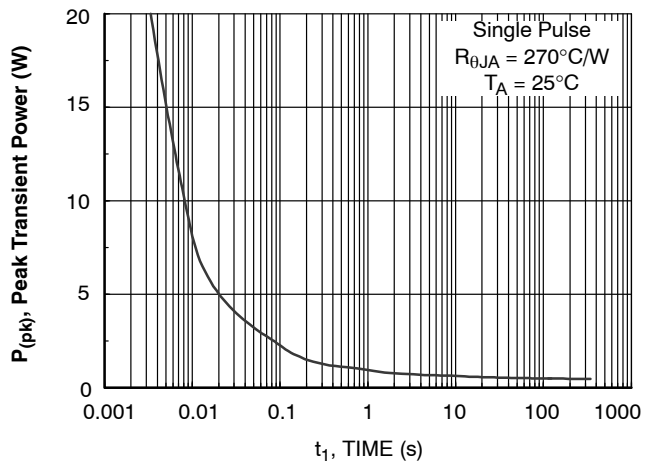


Figure 10. Single Pulse Maximum Power Dissipation

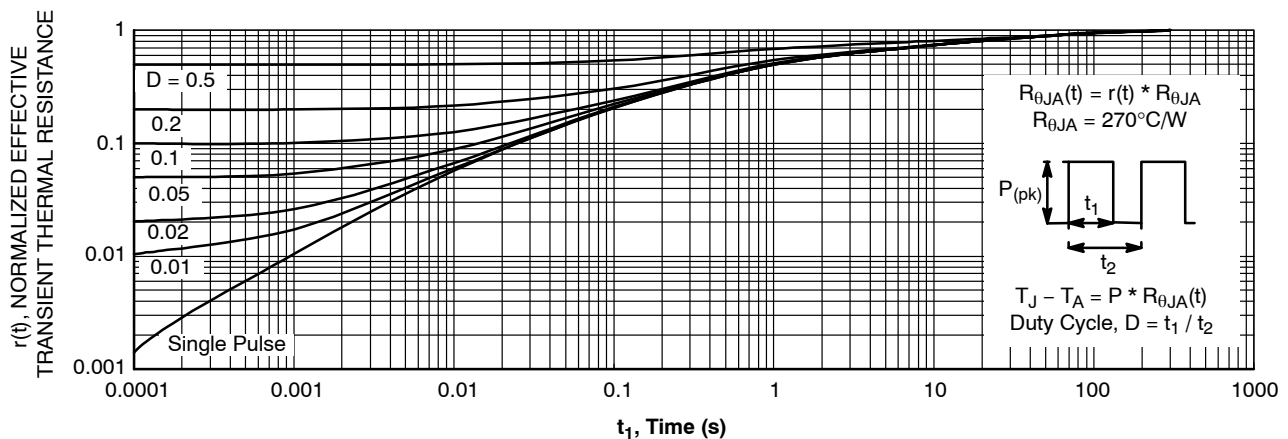
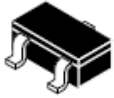


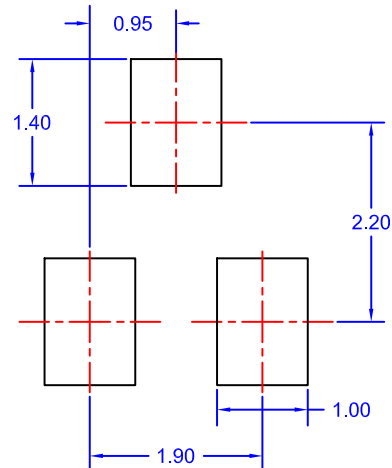
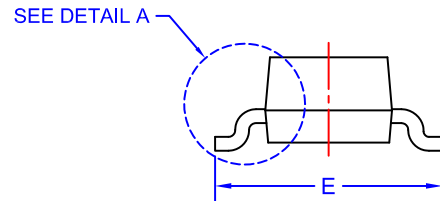
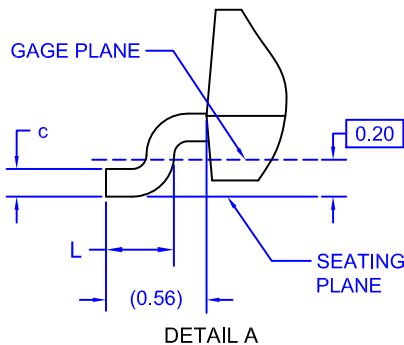
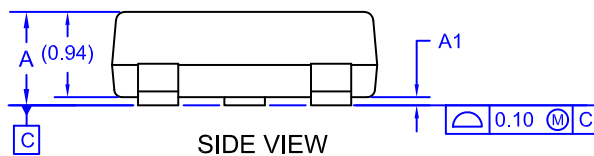
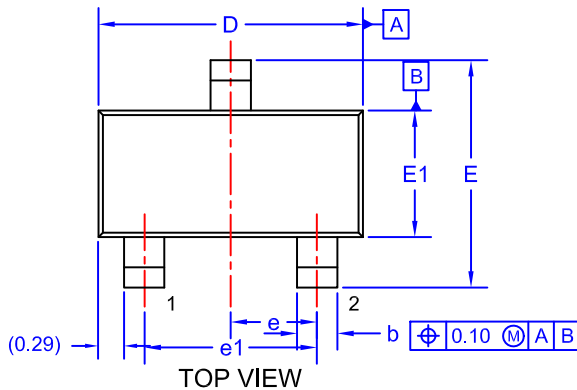
Figure 11. Transient Thermal Response Curve

Thermal characterization performed using the conditions described in Note 1b.
Transient thermal response will change depending on the circuit board design.

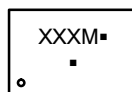
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SOT-23/SUPERSOT™ –23, 3 LEAD, 1.4x2.9
CASE 527AG
ISSUE A

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*FOR ADDITIONAL INFORMATION ON OUR Pb-FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE ON SEMICONDUCTOR SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

GENERIC MARKING DIAGRAM*


XXX = Specific Device Code
M = Month Code
▪ = Pb-Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.

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